

A star is born

The year was 1983. Return of the Jedi fittingly concluded the Star Wars series. It was also the year the word processor we're all using (mostly) was introduced under the name Multi-Tool Word and revolutionized text editing. It was around the same time that the world was introduced to a machine that behaved in a manner that was completely unlike Microsoft Word, but the change is brought about in the manufacturing process was as big as Microsoft Word's impact in its field

Whether you call it desktop fabrication or rapid prototyping or additive manufacturing what's being referred to is the process of 3D printing. Layer upon layer of a material is printed to create a finished product ready for sale that required no clay modeling, didn't come from a plaster negative and wasn't molded. It merely moved from the mock-up on your computer screen to the finished object on your desk.

But 'why all the hype?' you might wonder

Before we address the 'why' let's talk about the 'when'. While the 3D printing landscape seemed awfully quiet in the in the first decade of the new millennium, the number of 3D printer start ups, their rate of innovation and the number of products that are currently offered saw a sudden jump. This is attributed entirely to the expiration of key patents to what is known as Fused Deposition Modeling or FDM where a plastic is melted and layered on itself to produce an object.

It is with the expiry of this patent and a few others that start ups like Makerbot began making waves and 3D printing began to take off. Start ups like Makerbot, Printbot and even DIY projects like RepRap began offering 3D printers that took the quality of output to the level of industrial and commercial printers at a fraction of the cost. With this, in poured a whole bunch of 'Me Too' companies in to the 3D printing space.

A lot having been developed by the open source route too worked. The hardware, the software and the printable objects were free to be modified and improved by developers around the world this making more accessible to the masses kicking in to gear the 3D printing industrial revolution.

3D printing is a big deal because simply put, it has the potential to revolutionise mass production. Specialization of work and assembly lines could be completely phased out and one 3D printer could churn out item out after item.

3D printing could even phase out delivery of some products. Rather than having your latest product shipped, you could purchase or even download